CONTACT MANAGER

Introduction

Project Title : CONTACT MANAGER

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Project Overview

The **ConnectHub** is a backend web application designed to solve the common problem of scattered contact information across multiple platforms. It provides users with a centralized, organized system to store, search, and categorize their professional and personal contacts.

Key Objectives:

- Provide secure user authentication and data isolation.
- Enable efficient contact management through CRUD operations.
- Implement flexible categorization using tags.
- Offer fast search and filtering capabilities.
- Deliver a responsive, cross-device user experience.

Target Users:

- Freelancers and consultants managing client relationships.
- Small business owners tracking vendor and customer contacts.
- Students and young professionals building their network.
- Event organizers managing attendee and volunteer information.
- Anyone seeking a simple, privacy-focused contact management solution.

Architecture

Backend (Node.js + Express):

- **Controllers:** Business logic for authentication and CRUD operations.
- **Models:** Mongoose schemas defining data structure.
- **Routes:** API endpoint definitions with HTTP methods.
- **Middleware:** JWT verification, error handling, validation.
- **Configuration:** Database connection and environment variables.

Database (MongoDB):

- **Users Collection:** _id, name, email, password (hashed), timestamps.
- **Contacts Collection:** _id, userId (FK), name, email, phone, notes, tags[], timestamps.
- **Relationships:** Many-to-One (Contacts → User).
- **Indexes:** email (unique), userId, name, tags (for search).

PRE-REQUISITES

Software Requirements

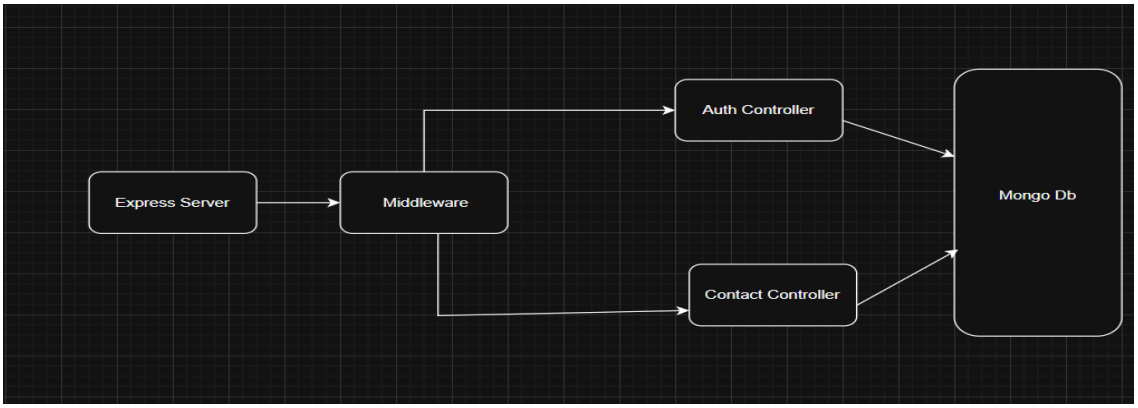
SOFTWARE	PURPOSE
Node.js	JavaScript runtime for backend
MongoDB	NoSQL database for data storage
npm	Package manager
Git	Version control system
Code Editor	VS Code

Knowledge Requirements

- JavaScript ES6+ syntax and concepts.
- Node.js and Express.js for RESTful API development.
- MongoDB and Mongoose ODM.
- JWT authentication and bcrypt password hashing.
- Git version control workflow.

PROJECT ARCHITECTURE

High-level architecture diagram illustrating the three-tier backend structure



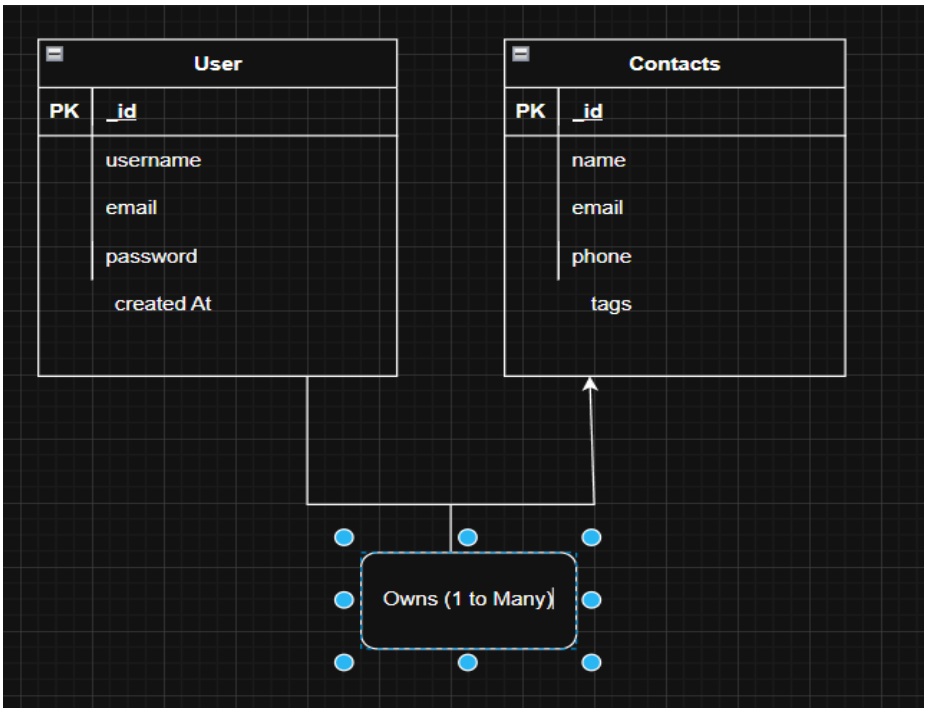
Node.js + Express Backend

- RESTful API architecture.
- JWT authentication middleware.
- MVC pattern (Models, Controllers, Routes).
- Input validation and error handling.

MongoDB Database

- Document-based NoSQL storage.
- Mongoose ODM for schema modelling.
- Two collections: Users and Contacts.
- Indexed fields for optimized queries.

ER Diagram



Roles and Responsibility

User -

Registration - Register on the application using the name, email , password.

Login - Login on the application using the email and registered password.

Search - Search for a contact using name , email or tags.

Add Contact - Add new contact - name , email , phone , notes , tags.

Edit Contact - Edit the existing contact.

Delete Contact - Delete any contact.

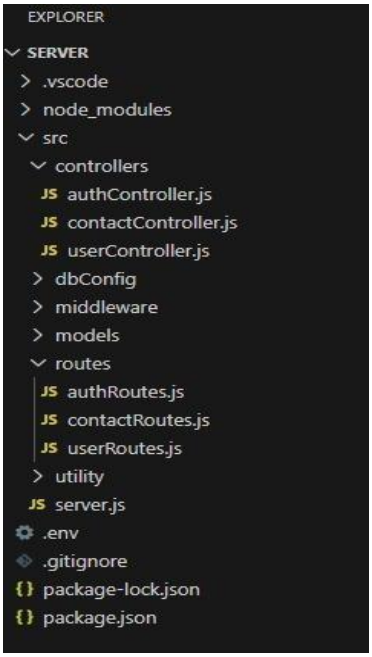
Running The Application:

Run: **npm run dev**

OR if not using nodemon: **node server.js**

Access : http://localhost:5000

Folder Structure



Backend Development

□ Setup express server

Create index.js file in the server (backend folder).
Create a .env file and define port number to access it globally.
Configure the server by adding cors, body-parser.

```
// Middleware
app.use(cors());
app.use(express.json());
app.use(express.urlencoded({ extended: false }));
```

□ User Authentication:

• Create routes and middleware for user registration, login, and

• Set up authentication middleware to protect routes that require user authentication.

```
import express from 'express';
const router = express.Router();
import { registerUser, loginUser, getMe } from '../controllers/authController.js';
import  authentication  from '../middleware/authMiddleware.js';
import { registerValidation, loginValidation } from '../utility/validators.js';
import validate from '../middleware/validation.js';

router.post('/register', registerValidation, validate, registerUser);
router.post('/login', loginValidation, validate, loginUser);
router.get('/me', authentication, getMe);

export default router;
```

□ Define API Routes:

• Create separate route files for different API functionalities such as user authentication and crud operation for contacts

• Implement route handlers using Express.js to handle requests and interact with the database.

```
router.get('/search', authentication, searchContacts);
router.route('/')
  .get(authentication, getContacts)
  .post(authentication, contactValidation, validate, createContact);

router.route('/:id')
  .get(authentication, getContactById)
  .put(authentication, updateContactValidation, validate, updateContact)
  .delete(authentication, deleteContact);
```

☐ Implement Data Models:

• Define Mongoose schemas for the different data entities like user and contact

• Create corresponding Mongoose models to interact with the MongoDB database.

• Implement CRUD operation(Create, Read, Update, Delete) for each model to perform database.

```
import mongoose from "mongoose";
import bcrypt from 'bcryptjs'
const userSchema = new mongoose.Schema(
  {
    name: {
      type: String,
      required: [true, 'Please add a name'],
      trim: true,
    },
    email: {
      type: String,
      required: [true, 'Please add an email'],
      unique: true,
      lowercase: true,
      trim: true,
      match: [
        /^\w+([\.-]?\w+)*@\w+([\.-]?\w+)*(\.\w{2,3})+$/,
        'Please add a valid email',
      ],
    },
    password: {
      type: String,
      required: [true, 'Please add a password'],
      minlength: 6,
      select: false,
    },
  },
  {
    timestamps: true,
  }
);
```

☐ Error Handling:

• Implement error handling middleware to catch and handle any errors that occur during the API requests.

• Return appropriate error responses with relevant error messages and HTTP status codes.

```
const errorHandler = (err, req, res, next) => {
  const statusCode = res.statusCode === 200 ? 500 : res.statusCode;

  res.status(statusCode).json({
    message: err.message,
    stack: process.env.NODE_ENV === 'production' ? null : err.stack,
  });
};

const notFound = (req, res, next) => {
  const error = new Error(`Not Found - ${req.originalUrl}`);
  res.status(404);
  next(error);
};

export { errorHandler, notFound };
```

Database Configuration and Setup

1. Configure MongoDB:

- Install Mongoose.
- Create database connection.
- Create Schemas & Models.

2. Connect database to backend:

Now, make sure the database is connected before performing any of the actions through the backend. The connection code looks similar to the one provided below.

```
const PORT = process.env.PORT || 6001;
mongoose.connect(process.env.MONGO_URL, {
    useNewUrlParser: true,
    useUnifiedTopology: true
}).then(()=>{


    server.listen(PORT, ()=>{
        console.log('Running @ ${PORT}');
    });


}).catch((err)=>{
    console.log("Error: ", err);
})
```

3. Configure Schema:

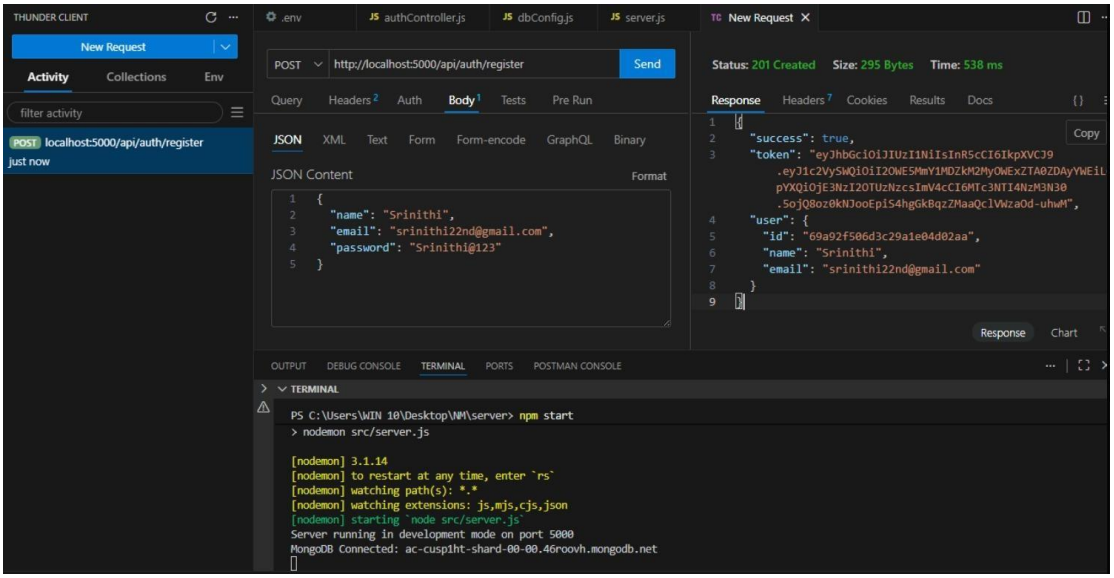
Firstly, configure the Schemas for MongoDB database, to store the data in such pattern. Use the data from the ER diagrams to create the schemas.

The schemas are looks like for the Application.

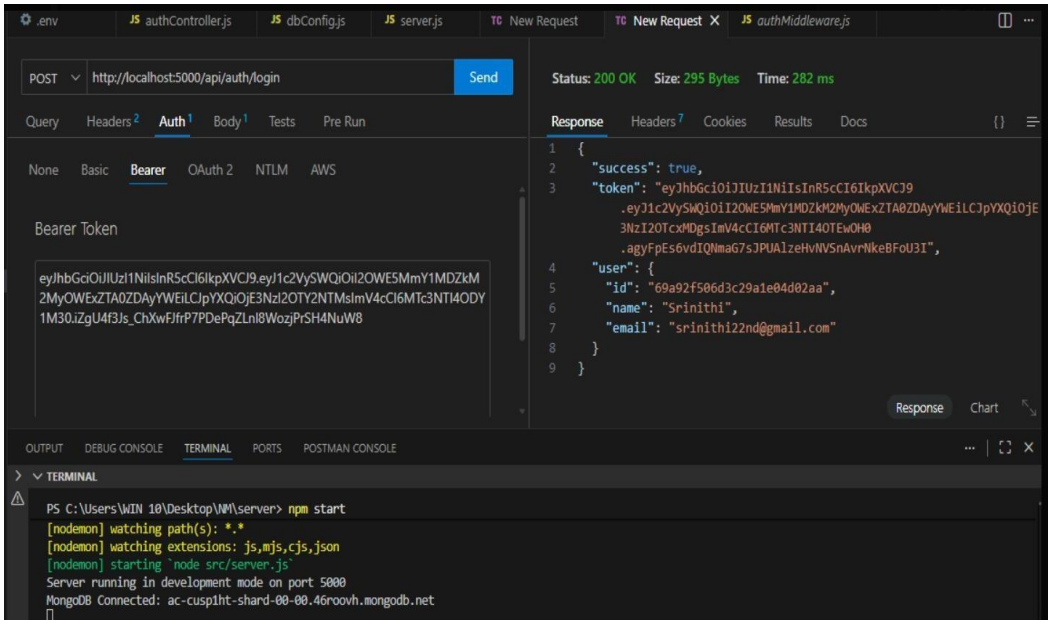
```
User.js
server > models > JS User.js > ...
import mongoose from 'mongoose';

const UserSchema = new mongoose.Schema({
    username:{
        type: String,
        require: true
    },
    email:{
        type: String,
        require: true,
        unique: true
    },
    password:{
        type: String,
        require: true
    },
});

const User = mongoose.model("users", UserSchema);
export default User;
```

Output screenshots

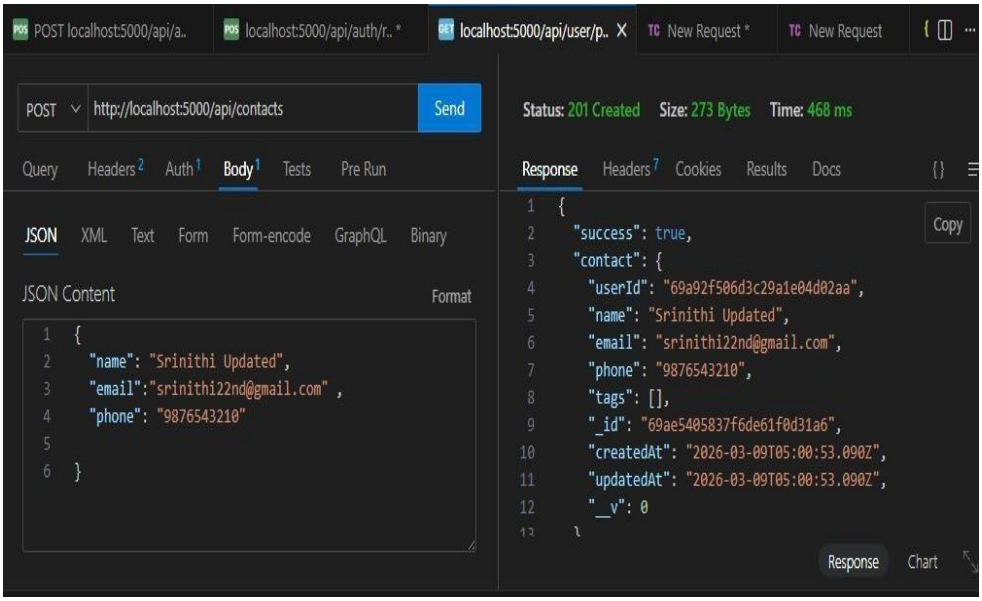
1. Register Api



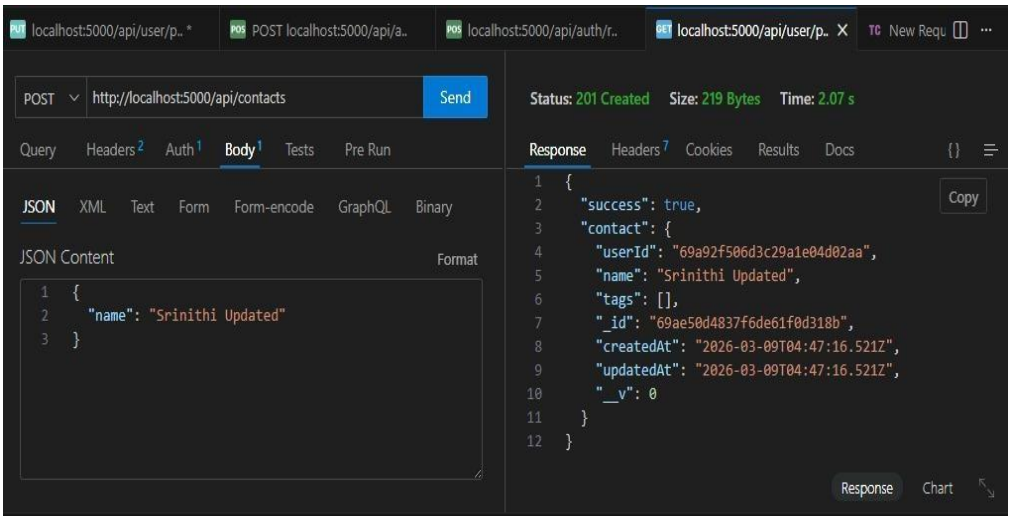
2. Login Api



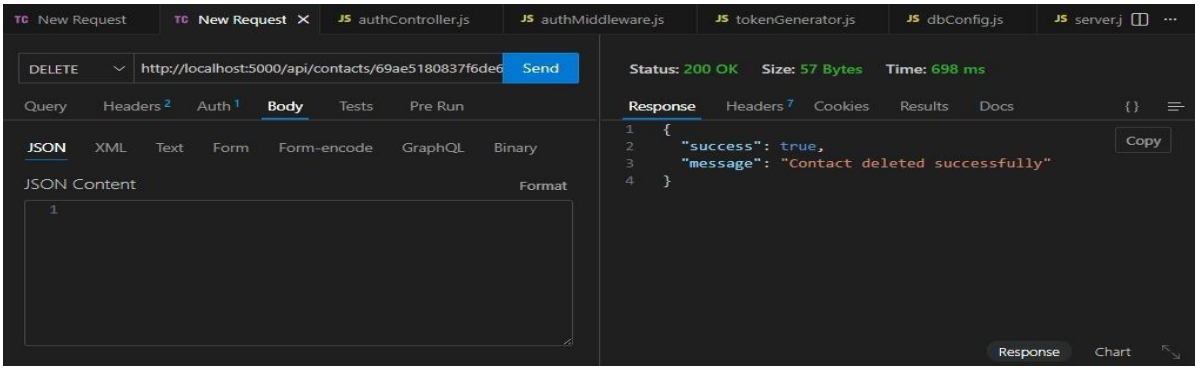
3. Create New Contact



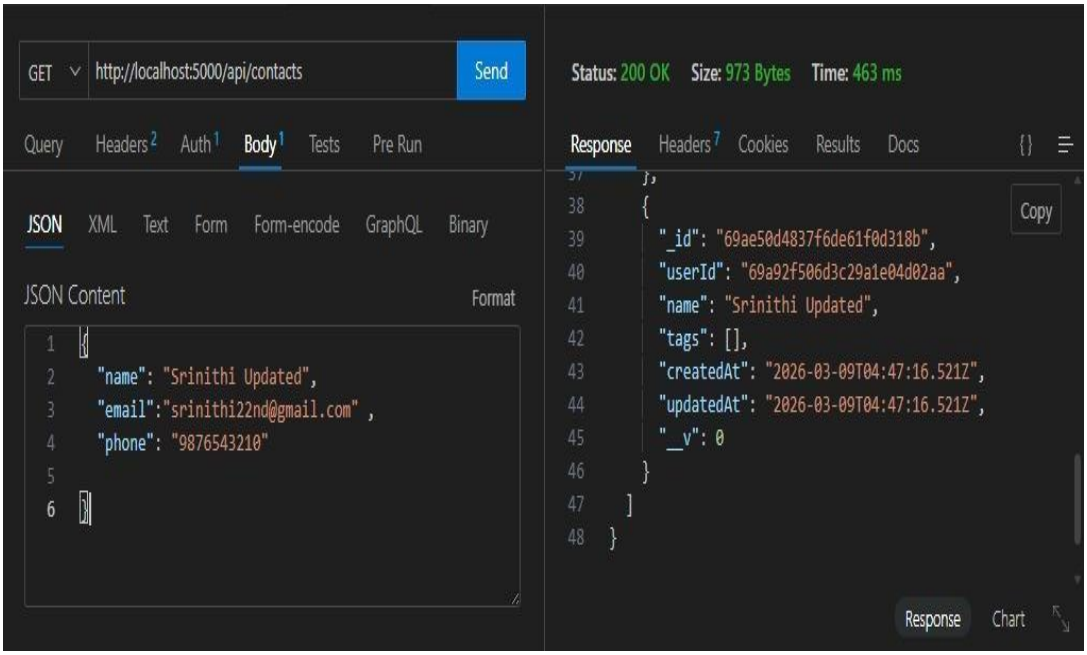
4. Update Contact



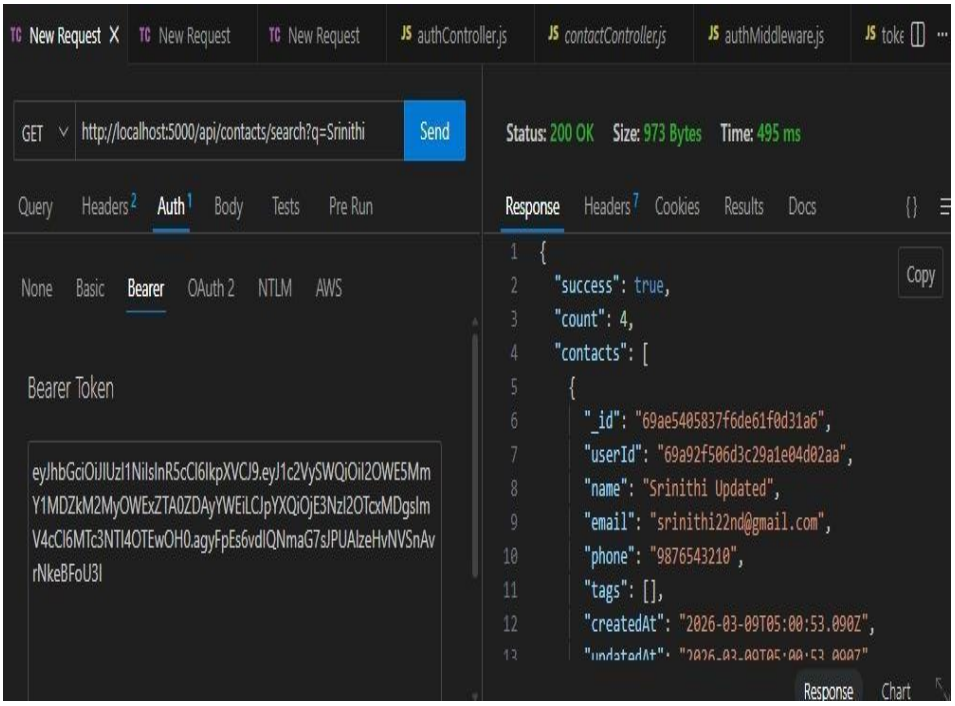
5. Delete Contact



6. Search Contact



7. Get The Current User



Demo link

https://drive.google.com/file/d/14U6-tgdeeCOddYmlnBLulhYsrJqp5qO7/view?usp=sharing